Negotiated Autonomy

Personal AI That Learns How We Delegate Spatial Work

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For discussion with Prof. Can Liu, City University of Hong Kong

An action that changes the agreement

A user asks an AI to improve a workspace while keeping the window workstation. The AI also moves a monitor the user expected to remain untouched. It understood the goal, but that understanding did not authorise every means of achieving it.

My research develops personal AI that connects conversation and perceived context with useful action. The question is how people delegate that action: “Move these chairs while I am away” grants latitude; “Do not touch that one” revises it. Such agreements change during work and across sessions.

My testable proposition is that at least 80% of grants in room-editing tasks have a scope expressible as a visible object set or region, an operation and a duration, without extra condition fields. Grants such as “only if the aisle stays clear” challenge that proposition. Their failures will identify what this compact representation leaves out.

Connections and the research claim

Reference-Based Manipulation [1] supplies Source, Anchor and Frame for locating permission; operations and duration complete the proposed scope record. EchoLadder [2] makes scene suggestions reversible. This project adds time-limited standing grants and a history of how those grants change across sessions. Orality [3] motivates treating granting, withholding and retracting permission as speech acts within ongoing thought.

Horvitz [4] establishes mixed initiative; automation levels [5] and human-AI guidelines [6] inform execution and correction. The new claim concerns the adequacy of a compact, spatially grounded permission representation and whether negotiating it costs less than editing explicit rules.

What the prototype learns

The desktop/XR prototype links speech and editor state to scene-editing tools. Floating models preview changes; gestures mark their scope. For each user and object-operation class, an estimator learns a default mode: suggest, preview, sandbox or commit. An approved standing grant lets that default govern repeated actions, reducing repeated authorisation. Scope expansion still requires a grant; silence supplies no positive training label. An inspectable history records changes, while speech or a palm gesture interrupts execution.

The questions concern how people express delegation, whether dialogue and learned defaults reduce its cost, and how agreements shift over repeated use.

Studies and measures

The first six months will produce a Wizard-of-Oz study with 14 participants and a taxonomy of authorisation speech acts. Spontaneous grants test coverage; targeted counterexamples diagnose failures. Interpretations are confirmed by participants, with ambiguous cases coded separately.

Thirty participants will then compare an editable rule panel, conversational authorisation, and conversational authorisation with learned defaults. All three share permission expressiveness and execution controls. The panel-versus-dialogue comparison tests how permission is expressed; adding learning tests whether defaults reduce repeated negotiation.

Delegation overhead counts authorisation utterances, gestures, rule edits and active time. Outcomes also include unwanted actions, completion and repair. Delayed autonomy drift means an action outside the latest explicit grant that remains unnoticed for 120 seconds of active inspection after becoming visible. Time to detection is recorded continuously; later regret and exploratory undo are separate. A six-week deployment with 12 participants will track permissions across sessions.

What I bring

Architectural training and CAVE installation experience ground my spatial interface work. Game Fellows provides a bounded companion prototype and a prepared Solo/Fellow comparison; human evaluation is planned. In my image-detection capstone, I reconstructed 900 decisions from frozen outputs and coefficients, exposing class-specific trade-offs hidden by overall improvement. That experience supports reconstructing an AI action against the permission in force when it occurred.

Selected References

[1] Yangyang He, Zhuangze Hou, Yonglin Chen and Can Liu. 2026. Reference-Based Manipulation: A Framework and Pipeline for Multimodal Spatial Reasoning. UIST 2026 (accepted). doi:10.1145/3830398.3830657

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[6] Saleema Amershi et al. 2019. Guidelines for Human-AI Interaction. CHI 2019. doi:10.1145/3290605.3300233